

03.01 Systems of Linear Equations

Common problems and solution methods

Rule guide: recognize the task first

Recognize	Apply these rules
Linear Relationships And Constraints	Choose the representation that matches the wording: equation, inequality, system, or objective function.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

Example 1: Identify the model

Problem

$$2x + y = 7$$

How to solve it

Write the known quantities and restrictions.

$$y = mx + b$$

Check the result against the original statement.

Example 2: Apply the key rule

Problem

$$y > -2x + 3$$

How to solve it

Write the known quantities and restrictions.

$$ax + by = c$$

Check the result against the original statement.

Example 3: Check the result

Problem

$$P = 4x + 3y$$

How to solve it

Write the known quantities and restrictions.

$$P = ax + by$$

Check the result against the original statement.